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Discussion Agenda 11/16/2012

Below is a discussion agenda in an approximate order of importance for the first Web Conference of the Project AGORA (Nov.16, 9am PST/6pm CET).

This document was first reviewed and discussed by the **steering committee of the Project** (Abel, Gnedin, Madau, Mayer, Primack, Teyssier, Wadsley, and others) on Nov. 13, and then by the whole collaboration on Nov. 16.

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1. Progress Report (as of 11/16/2012)

The following table summarizes the progress made so far in each Working Group (updated by Working Group leaders based on the [action items](#) announced on Oct. 6).

	WORKING GROUP	PROGRESS
I	Common Physics and Introduction to Project	Common physics package (chiefly the routine that calculates the cooling rate as a function of temperature, metallicity, etc.) is getting close to be distributed thanks in large part to the hard work by Britton Smith, Sijing Shen, Mike Kuhlen, and Matthew Turk. In addition, the flagship paper is being drafted by Joel Primack and Piero Madau to introduce the rationale of the Project. We intend to include a few snapshots from the fiducial runs (isolated+cosmological; see Part 5 below). The flagship paper will hopefully be circulated by the end of November, and submitted for publication by the end of 2012.
II	Common ICs: Isolated	Initial conditions have been generated using a older version of Volker Springel's code MakeDisk for a 1e12 Msun halo with an old stellar disk and a gas disk with 20% gas fraction. The ICs exist in low, medium and high resolution and are available at the following website: http://kicp.uchicago.edu/~agertz/disk/ . The LOW and MED initial conditions have been run with many codes already: ART, Gadget, Ramses, Gasoline, and SPHS. It seems they give rise to the formation of a relatively stable disk, depending on the details of the various feedback implementations.
III	Common ICs: Cosmological	The MUSIC configurations files for all four masses (1e10 – 1e13 Msun) with quiescent merger histories are up on the Workspace, so we can kickstart the project with these ICs at any time. Anybody is welcome to test the ICs in a dark matter only runs, and report any problem. WG III is experiencing issues with the ICs with violent histories (hard to find a candidate that experienced a violent merger history, yet fits the isolation criteria); however, our group is continuously working on this front.
IV	Common Analysis	Work is ongoing on yt. Currently, for quantitative analysis and visualization, codes RAMSES, Enzo, Nyx are fully supported. ART is in progress, with NMSU ART a near-term target. Particle-only data can be read with fast load-on-demand and analyzed quantitatively, and image-based visualization is in progress. High-priority has been assigned to particle-based data, and several venues for short term solutions as well as longer term solutions are being explored.
V	Isolated Galaxies and Subgrid Physics	We have run some preliminary simulations of the isolated disk IC for a Milky Way size halo with as many as 5 different codes. Results are qualitatively similar, but quantitatively, both code types and resolution seem to matter. Next step is to define some common set of diagnostics and start to explore the effect of subgrid physics.
VI	Dwarf Galaxies in Cosmological Simulations	1e10 Msun quiescent and merger candidates selected. Zoom-in dark-matter only tests using Gadget-2 for the quiescent halo finished and MUSIC .conf file to generate ICs for this halo is already available. Currently confirming the participation of the different codes representatives that originally show interested in the working group starting dark-matter only runs with them. Progress of WG VI has been slower than expected because dark-matter only tests took more time than originally thought. Once the tests for the other halo finished, its MUSIC .conf file will also be posted. A mailing list has been created for all people participating in the working group. Workspace updated with all information.
VII	Dark Matter	No major progress to report. We have established group leadership, updated the WG page including a list of scientific questions, and ensured that we have representation from all major codes (no Arepo). Some DM-only simulations have been started. Looking forward to getting our hands dirty with analysis once more simulations have progressed.

VIII	Satellite Galaxies	There is no science progress to report, but action items were performed. The workspace for this group is being developed with our science agenda, and a mailing list has been created for the participants in this group. We are working to make sure every code is represented, and would like members of the Gadget, HART and Ramses codes to join this working group.
IX	Characteristics of Cosmological Galaxies	The action items were performed. The (dual) leadership of the team is confirmed, all codes and major analysis tools (yt, Sunrise, etc) are well represented, with the exception of Nyx. The tentative science plan is laid out on the team's workspace.
X	Outflows	Action items were performed. The lists of science questions and code representatives are posted to the workspace. The leadership of the group is confirmed. A mailing list is created for group members to share progress and ideas. All participating codes in the Project are represented in WG X. No cosmological hydrodynamic simulation or analysis is performed at this stage. The tipsy format initial condition for $1e10$ and $1e11$ Msun halos are generated and currently under test.
XI	High-redshift Galaxies	The action items were performed. Most of the codes are well represented. Only SPHS and Nyx are missing. Leadership of the group is confirmed. A list of specific scientific questions, related to high-z galaxies is being compiled.
XII	Interstellar Medium	There is no progress in analysis or simulations to report. Action items were performed: codes except for Nyx, and Arepo are represented; leadership is confirmed; some priority science questions and a tentative simulation agenda have been posted to the workspace.
XIII	Black Hole Accretion and Feedback	The action items were performed. Codes represented are Gadget, Gadget-SPHS, RAMSES, ART, Enzo. Participation from anyone representing an additional SPH code (i.e. Gasoline) would be greatly appreciated. SMBH sub-grid modelling strategy is being discussed.

[2. Coordination Between Working Groups on Choosing Simulation Campaigns](#)

This is something we may want to discuss before it is too late. All 9 science Working Groups do not have to run 9 different simulations, because some of them could be more or less similar, and differ only by analysis. It would be great if Working Groups start talking to each other for a coordinated simulation effort.

Indeed, one priority should be to take advantage of the common physics and common ICs by running at least one set of simulations with common "basic" star formation and feedback physics that can be implemented in all codes. Those simulations could be analyzed, as a baseline, by all working groups. For more realistic simulations, Working Group leaders could petition for specific sets of runs; possibly doing this through a new workspace or via a future smaller telecon that includes simulators. The resulting choices would be motivated by the specific goals of individual science groups, but performed based on what is interesting and practical for all of the Project.

Therefore, **the steering committee recommends** the following working group leaders to discuss and choose "simulation campaigns", at least for the first stages of their simulations.

	INITIAL CONDITION TYPES	WORKING GROUPS
1	Isolated IC	Working Group V – Isolated Galaxies and Subgrid Physics Working Group XIII – Black Hole Accretion and Feedback
2	Cosmological IC : Start with DM Only	Working Group VI – Dwarf Galaxies in Cosmological Simulations Working Group VII – Dark Matter
3	Cosmological IC : Start with Dwarfs	Working Group VIII – Satellite Galaxies Working Group IX – Characteristics of Cosmological Galaxies Working Group X – Outflows Working Group XI – High-redshift Galaxies Working Group XII – Interstellar Medium

[3. Oversight and Publication Policies](#)

The AGORA steering committee proposes the following policies for the authorship and publication of the papers resulting from the Project AGORA.

PAPER / ITEM	POLICY FOR AUTHORSHIP AND PUBLICATION
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1	The Flagship paper of the Project	<i>[Option 1]</i> All the people who have signed up in the collaboration, regardless of their contribution at the point of the submission of the article. <i>[Option 2 – chosen by the online poll on Nov. 16–20, 2012]</i> All the people who have made meaningful contributions to the Project (not just to the flagship paper) up to the point of the submission of the article. The steering committee makes judgements on the contribution and decides the authorship. <i>[Option 3]</i> Only the people who have made significant contributions to the text of the flagship paper (on project rationale, ICs, physics, working groups, etc). But there is an additional tag "for The AGORA Collaboration" at the end.
2	The primary paper of each Working Group	The leader of the Working Group functions as a mediator to decide the authorship of the paper (and its order) in consultation with the steering committee.
3	Any other paper (e.g. a spin-off or a sub-project in any Working Group)	The person who first proposed and initiated such project functions as a mediator to decide the authorship of the paper (and its order) in consultation with the steering committee.
4	Journals and publication charges	The lead author chooses which journal to submit: ApJ, MNRAS, New Astronomy, etc. Should the authors of the flagship paper and the paper(s) to introduce ICs choose to submit the papers to ApJ(S), they would receive financial support from the UC–HiPACC for publication charges. No financial support on other papers from UC–HiPACC.

In addition, **the steering committee will act as a publication board** when a paper resulting from the Project AGORA is ready to be submitted to an academic journal. It will provide editorial advice on the article, and make sure that due credits are given to appropriate authors. Each member of the steering committee has also agreed to provide general guidance and oversight to 1–2 science-oriented Working Groups from the conception of their simulation campaigns.

	MEMBER	WORKING GROUPS
1	Nick Gnedin	Working Group XII – Interstellar Medium
2	Piero Madau	Working Group VII – Dark Matter Working Group X – Outflows
3	Lucio Mayer	Working Group VIII – Satellite Galaxies Working Group IX – Characteristics of Cosmological Galaxies
4	Joel Primack	Working Group VI – Dwarf Galaxies in Cosmological Simulations Working Group XI – High-redshift Galaxies
5	Romain Teyssier	Working Group V – Isolated Galaxies and Subgrid Physics Working Group XIII – Black Hole Accretion and Feedback
6	James Wadsley	Working Group X – Outflows

4. Milestones 2012-2013

These are the milestones expected to be met in the next ~12 months.

	TIMELINE	MILESTONE
1	By the end of December, 2012	– The flagship paper circulated and submitted – Various simulation campaigns selected and launched
2	Early 2013	– The second Web Conference
3	Spring of 2013	– Paper(s) submitted by Working Groups II and III describing the ICs – Common ICs and common physics package become public – The peta-byte data storage system at UCSC comes online (will store publicly-readable simulation data just like the Millennium consortium)

		did; password-protected write access by Working Group leaders; the data of the Working Group becomes public as soon as their publication is submitted)
4	Summer of 2013	- First science papers by Working Groups V (isolated galaxy) and VI (DM-only comparison of dwarfs)
5	Aug. 16–18, 2013 (tentatively)	- The second AGORA Workshop at UCSC right after the annual Santa Cruz Galaxy Workshop 2013
6	Fall and Winter of 2013	- The next wave of science papers by Working Groups VII – XIII

5. DM-only Run of A $\sim 1e10$ (or $\sim 1e11$) M_{sun} Halo for the Flagship Paper

In the flagship paper, Working Group I intends to include a panel of snapshots of the fiducial runs (isolated + cosmological) performed by each of the participating codes. In particular, we need a volunteer from each of the participating codes who can run a quick dark matter only cosmological simulation (described below) and generate a few images and profiles.

- DM-only simulation with a 10^{10} (or 10^{11}) M_{sun} halo with a quiescent merger history, <1 kpc/h resolution
- Needed: particle projection at $z=0$, merger history, halo mass, dark matter density profile, etc.
- **The details of this fiducial run including the target halo mass and resolution will be decided in the coming weeks**
- See [this page](#) for the continuing discussion on this topic in case you are interested

	CODE	VOLUNTEER
1	ART-Chicago	Sam Leitner
2	ART-NMSU	Anatoly Klypin
3	ENZO	Ji-hoon Kim
4	GADGET3(-SPHS)	Alex Hobbs, Justin Read
5	GADGET3-adaptive force softening	Alex Hobbs, Justin Read (in consultation with F. Iannuzzi)
6	GASOLINE	Sijing Shen
7	Nyx	Jan Engels, Wolfram Schmidt
8	PKDGRAV2	Mike Kuhlen
9	RAMSES	Oliver Hahn, Romain Teyssier

6. New Participants

There has been a number of new participants in the project; please have a look at the up-to-date participants list at [Useful Info](#). Notably, Nyx code is now being represented by Dr. Wolfram Schmidt and his colleagues at the University of Gottingen. **Working group leaders are encouraged to have a look at the most recent list of participants who have signed up for their respective groups at [Project Details](#).** The leaders are also asked to be very inclusive of all group members at every step of the way, especially graduate students.

7. New Website

The steering committee is in the process of purchasing a new domain (www.agorasimulations.org, tentatively), which will replace the current public front, sites.google.com/site/santacruzcomparisonproject/.

